

Project Executable Files

Date	01 July 2026
Team ID	
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA (<i>Uses CSV dataset instead of a database</i>)
5	Environment configuration file (.env.example or similar)	Yes
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	No
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

```

OptiCrop/
├── app.py                # Flask application (all routes)
├── requirements.txt
├── README.md
├── data/
│   └── crop_data.csv    # Generated training dataset (2,760 records, 23 crops)
├── model/
│   ├── train_models.py  # ML training pipeline
│   ├── crop_model.pkl   # Best model (Random Forest, 92.57%)
│   ├── scaler.pkl       # StandardScaler
│   ├── label_encoder.pkl # LabelEncoder (23 crops)
│   ├── kmeans.pkl       # K-Means clustering model
│   └── model_meta.json  # Metrics, feature importances, crop profiles
└── templates/
    ├── base.html        # Shared layout & navbar
    ├── index.html        # Home page (Scenario overview)
    ├── predict.html      # Scenario 1 – Crop Recommendation
    ├── suitability.html   # Scenario 2 – Suitability Assessment
    └── research.html     # Scenario 3 – Research & Analytics
  
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://opti-crop-21mj28erl-s-sai-charans-projects.vercel.app/
Login Credentials (Demo)	Not Applicable (No login required)
Platform / Hosting Provider	Vercel (Frontend Hosting), Flask (Backend API), GitHub (Source Code Repository)
Repository Link	https://github.com/saicharan065/OPTICROP
Demo Video Link	https://youtu.be/4sT0auZbJ2k

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

Pre-requisites: Install Python 3.x, Node.js, npm, Visual Studio Code, Git, and the required project dependencies. Ensure that an internet connection is available for accessing the deployed application or downloading the necessary packages.

Step 1: Clone or download the OptiCrop project from the GitHub repository.

Step 2: Install the required frontend and backend dependencies using npm install and pip install -r requirements.txt.

Step 3: Start the Flask backend by running python app.py.

Step 4: Start the React application using npm run dev or access the deployed application through the Vercel URL.

Step 5: Open the application in a web browser, enter the soil nutrient values (Nitrogen, Phosphorous, Potassium) along with temperature, humidity, pH, and rainfall details, and submit the form.

Step 6: View the recommended crop, confidence score, top crop predictions, farming tips, and crop suitability analysis with parameter gap suggestions.

Step 5: Known Issues / Limitations

S. No	Known Issue / Limitation	Workaround / Status
1	The application uses a trained dataset and does not use real-time weather or soil sensor data.	Planned to integrate live weather APIs and IoT-based soil sensors in future versions.
2	Crop recommendations depend on the quality and accuracy of the input values provided by the user.	Users should enter correct soil test values to obtain accurate recommendations.
3	The current version focuses only on crop recommendation and crop suitability analysis.	Future updates will include fertilizer recommendation, crop disease prediction, and yield estimation.